Dataset Guide

Hotel Booking Data Analysis and Business Intelligence Dashboard

1. Files Supplied

File	Rows	Role	Description
hotel_bookings.csv	34,662	Fact table	One row per booking (includes 150 deliberate duplicate rows)
dim_hotel.csv	2	Dimension	Hotel name, type, location, total rooms, star rating
dim_room_type.csv	4	Dimension	Room type code and description
dim_country.csv	10	Dimension	Country code, name and region (for the map)

Overview: 2 hotels (Chennai City Hotel, 60 rooms; ECR Beach Resort, 40 rooms), arrivals from 1 Jan 2022 to 31 Dec 2024, about 34,500 unique bookings, prices in INR. About 31% of bookings are cancelled. Bookings are made up to about 15 months ahead of arrival, so booking_date starts in late 2020.

2. Relationships (for the Data Modeling stage)

From (many)	To (one)	Key
hotel_bookings[hotel_id]	dim_hotel[hotel_id]	hotel_id
hotel_bookings[room_type_reserved]	dim_room_type[room_type]	room type code
hotel_bookings[country]	dim_country[country_code]	ISO 3-letter code
hotel_bookings[arrival_date]	Date table[Date]	Create the date table with DAX

3. Data Dictionary: hotel_bookings.csv

Column	Meaning
booking_id	Unique booking number (BK000001 ...)
hotel_id / hotel	H01 Chennai City Hotel, H02 ECR Beach Resort
booking_date	Date the booking was made
arrival_date / departure_date	Check-in and check-out dates
lead_time	Days between booking date and arrival date
stays_in_weekend_nights	Nights falling on Friday or Saturday
stays_in_week_nights	Remaining nights of the stay

Column	Meaning
adults / children / babies	Guests of each type
meal_plan	RO room only, BB breakfast, HB half board, FB full board
country	Guest country, ISO 3-letter code
market_segment	Online TA, Offline TA/TO, Direct, Corporate, Groups, Complementary
distribution_channel	TA/TO, Direct, Corporate, GDS
customer_type	Transient, Transient-Party, Contract, Group
is_repeated_guest	1 repeat guest, 0 new guest
previous_cancellations	Earlier bookings this guest cancelled
previous_bookings_not_canceled	Earlier bookings this guest did not cancel
room_type_reserved / room_type_assigned	A Standard, B Deluxe, C Premium, D Suite
booking_changes	Number of amendments to the booking
deposit_type	No Deposit, Refundable, Non Refund
waiting_list_days	Days on the waiting list before confirmation
adr_inr	Average daily rate in INR (price per room per night)
parking_spaces	Parking spaces requested (0 or 1)
special_requests	Number of special requests (0 to 4)
is_canceled	1 cancelled or no-show, 0 stayed
reservation_status	Check-Out, Canceled, No-Show
reservation_status_date	Check-out date, or the date of cancellation

4. Deliberate Data Quality Issues (to fix in Stage 3)

Real data is never perfect, so these issues were planted on purpose. Finding and fixing them is the Data Cleaning stage.

Issue	Count	Fix in Power Query
Blank children values	300	Replace null with 0
Blank country values	about 500	Replace null with "Unknown"
Zero-guest bookings (adults, children, babies all 0)	60	Filter out
Negative adr_inr	25	Remove rows
Extreme adr_inr values (8 to 14 times normal)	20	Review, then remove
Exact duplicate rows	150	Remove duplicates (should leave 34,512 rows before other removals)

Tip: after removing duplicates, booking_id should be unique. Use it to check your work.

5. Patterns Built Into the Data

The data follows realistic hotel behaviour, so your dashboard will show meaningful insights. Do not treat the list below as the answer. Build the visuals first and see whether you find these patterns yourself.

- Seasonality: peaks around December and January; slower months in mid-year (city) and in the monsoon months (resort).
- Cancellations rise strongly with lead time.
- Online travel agent and group bookings cancel far more than Direct and Corporate bookings.
- Non-refundable bookings cancel much less than flexible ones.
- Repeat guests cancel less.
- The resort has a higher ADR and a higher cancellation rate than the city hotel.
- Domestic (Indian) guests dominate; the resort draws a larger international share.
- Occupancy is capped by room count: neither hotel exceeds its rooms on any night.